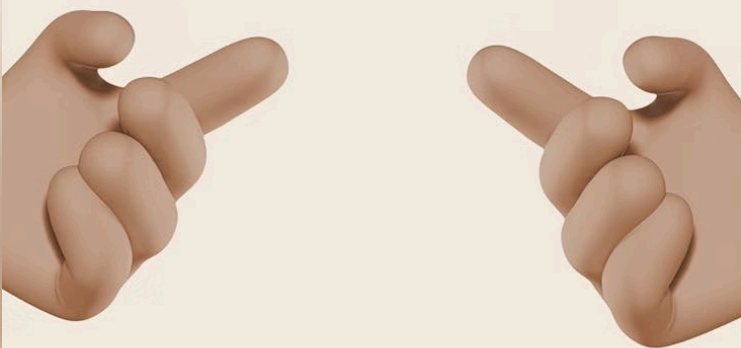


SIGNBRIDGE

REAL-TIME BIDIRECTIONAL SIGN LANGUAGE TRANSLATOR

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2025-27



466M

**PEOPLE LIVE WITH DISABLING
HEARING LOSS WORLDWIDE**

THE PROBLEM WE SET OUT TO SOLVE

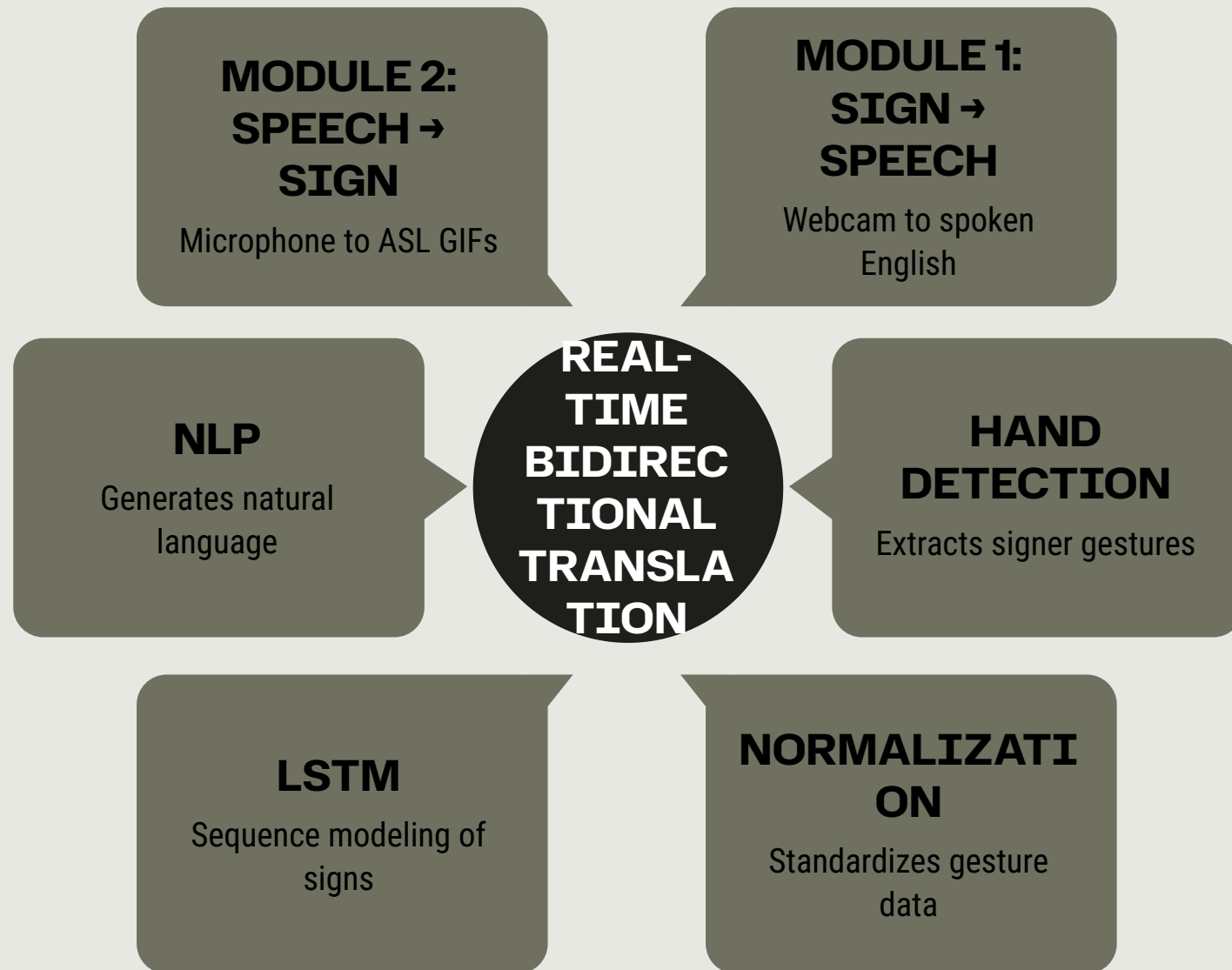
Most sign language technology only works one way – sign to text, or text to sign. Never both at once.

| Imagine a live concert interpreter – but automated, running on your laptop, and working in both directions simultaneously.

What we built: a real-time, two-way ASL translator. No GPU required. No special hardware. Just a webcam, a browser, and a conversation.

HOW THE SYSTEM WORKS

Two modules run simultaneously, connected by a live web layer.



Flask-SocketIO streams everything live to the browser. Two people on the same WiFi can use SignBridge from their own devices with zero setup required.

DATA COLLECTION & NORMALIZATION

TRAINING DATA

A custom script records **30 frames per sign**, triggered by a SPACE key press. 32 classes · 30 sequences each · Recorded on a MacBook webcam.

THE DOMAIN GAP PROBLEM

Train at 50 cm, deploy at 100 cm — predictions collapse. Every sign language ML system struggles when camera distance changes.

OUR FIX: WRIST-ORIGIN NORMALIZATION

STEP 1 — CENTER

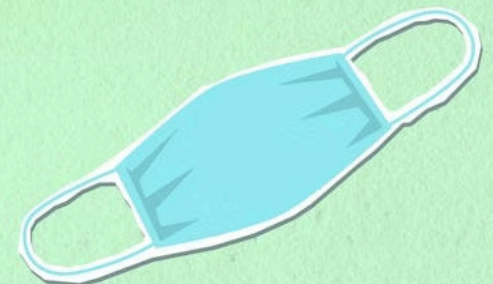
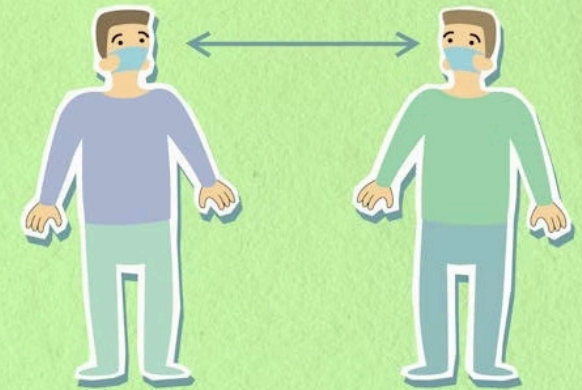
Subtract wrist position from all 21 hand landmarks.

STEP 2 — SCALE

Divide by maximum landmark distance to remove scale.

RESULT

Same feature vector — regardless of camera distance.



WHY LSTM? THE MODEL CHOICE

ARCHITECTURE COMPARISON

Model	Issue
Plain RNN	Forgets earlier frames before the sequence ends
Transformer	Needs far more data; heavier CPU cost
LSTM ✓	Handles 30-frame sequences well. Fast. Works on small datasets.

Architecture: Input (30 frames, 126 features) → LSTM 128 → LSTM 64 → Dense → Softmax over 32 classes

θ1

HAND VISIBLE?

θ3

CONFIDENCE ABOVE THRESHOLD?

θ2

HAND STILL?

θ4

TOP PREDICTION CLEARLY LEADS?

INFERENCE SAFEGUARDS

10 consecutive agreeing frames before committing. 3-second cooldown between detections. T5 gloss converts letters into natural sentences.

SPEECH TO SIGN

HOW IT WORKS

The user speaks or types a sentence. Each word is matched to a GIF from the ASL library. Unknown words are fingerspelled letter by letter.

WHY GIFS?

Simple, universally supported, lightweight. No video codec, no buffering — just clear, looping sign demonstrations.

PUSH-TO-TALK DESIGN

Silence detection introduced **400ms delays** and misfired on background noise. A single keypress to start and stop eliminated both problems entirely.

IN THE BROWSER

The Web Speech API runs with continuous listening and interim results — words appear as you speak, GIFs start immediately on confirmation.



HANDLING HARD CASES

Similar hand shapes are the hardest part of ASL recognition. We built a **cosine angle discriminator** – invoked only when the model is uncertain.

ANGULAR SEPARATION BY CONFUSED PAIR

Pair	Angular Gap
V vs Z	22.6°
N vs J	12.4°
O vs E	6.9°
B vs C	5.2°

Every confused pair is separable. The discriminator has not misfired on a validated input.

Sign J – The One Honest Limitation

J is a motion sign, tracing a path through the air. Our model catches it roughly two-thirds of the time. An LSTM-GRU hybrid or optical flow would handle this properly. It's on the roadmap.

LLM VERIFICATION & DEPLOYMENT

GROUNDING SIGNS IN REAL SOURCES

θ1

RETRIEVE

Look up the sign in a curated knowledge base sourced from ASL University and Gallaudet University.

θ2

AUGMENT

Inject the authentic description into a prompt for Llama3.

θ3

GENERATE

Llama3 returns a verdict – Verified or Uncertain – fully on-device via Ollama. No cloud. No API key.

THE WEB INTERFACE – FOUR TABS

SIGN TO SPEECH

Live camera, sign buffer, sentence conversion

SPEECH TO SIGN

Mic input, GIF playback queue

LIVE NETWORK

Create a room, share a code, peer joins instantly

HISTORY

Save and export translated conversations

To run: `python app.py` → open browser → done.

WHAT'S NEXT

WHERE WE ARE

A working bidirectional ASL translator. Browser-based. CPU-only. Verified against real ASL sources.

THE ROADMAP



MOBILE

TensorFlow Lite for iOS and Android



VOCABULARY

Transfer learning from WLASL – 32 to 500+ signs



MOTION SIGNS

LSTM-GRU hybrid to properly handle J and Z



MULTI-SIGNER & PEER VIDEO

Per-signer fine-tuning · WebRTC browser calls



The communication barrier between deaf and hearing people isn't a hardware problem. The hardware already exists. This is an engineering and prioritization problem – and we think it's one worth solving.